



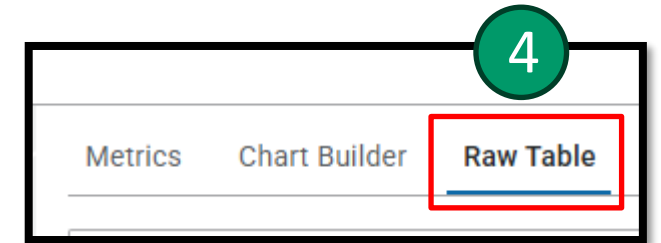
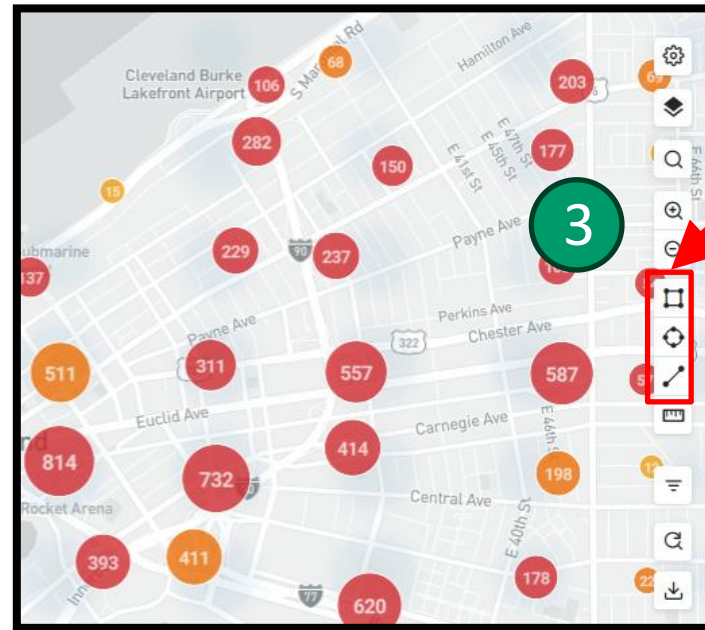
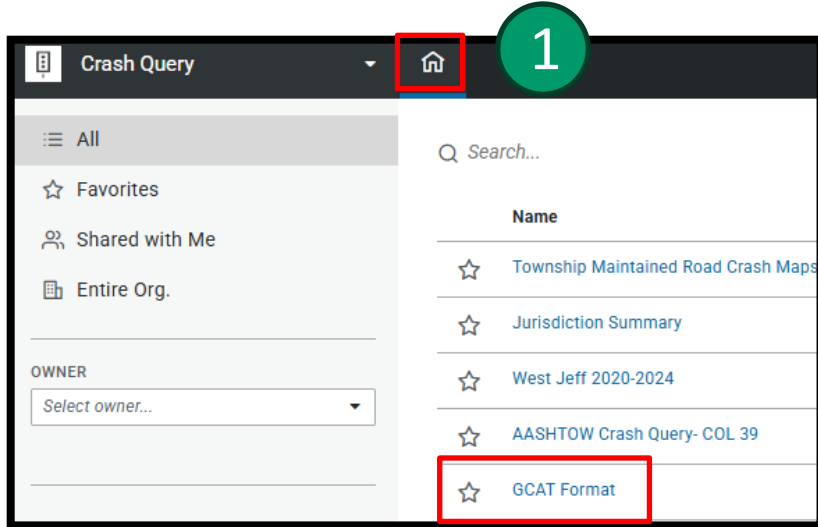
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CAM TOOL INSTRUCTIONS

STEP 1

- Export a crash data csv file from AASHTOWare Safety using *GCAT Format*.



The screenshot shows the 'Download' button. The 'Download' button is highlighted with a red box and circled with a green '5'.

☒	202034132
☒	202034191
☒	202037030

Crash Hide/Show Columns Download



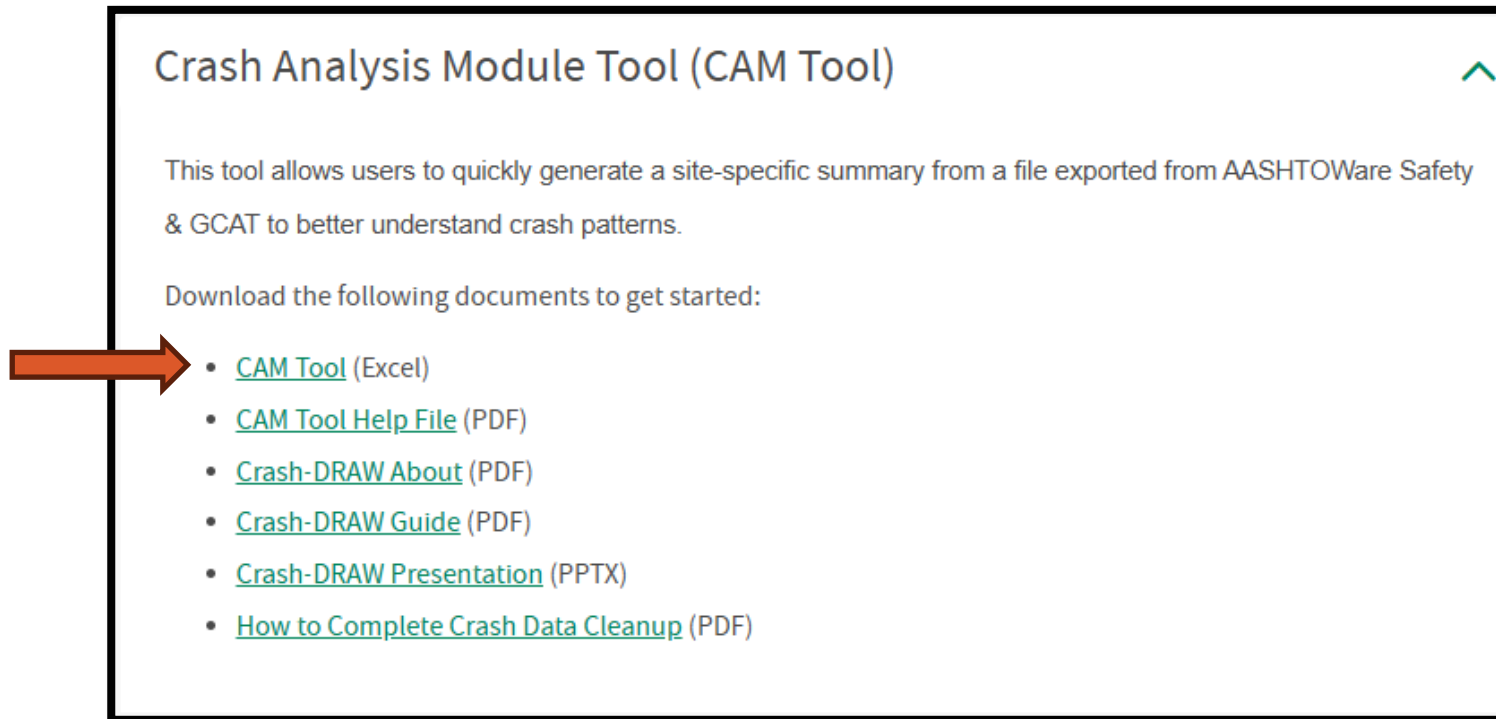
NEED ACCESS?

- Request **AASHTOWare Safety** Access Here:
https://odot.formstack.com/forms/aashtoware_safety_access_request
- Site Login: [AASHTOWare Safety - Crash Query](#)



STEP 2

- Download the **CAM Tool** from ODOT's Crash Analysis Tools Website.



Crash Analysis Module Tool (CAM Tool) 

This tool allows users to quickly generate a site-specific summary from a file exported from AASHTOWare Safety & GCAT to better understand crash patterns.

Download the following documents to get started:

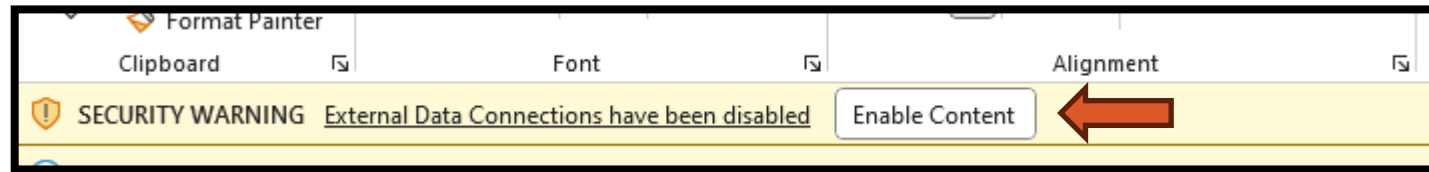
- [CAM Tool](#) (Excel)
- [CAM Tool Help File](#) (PDF)
- [Crash-DRAW About](#) (PDF)
- [Crash-DRAW Guide](#) (PDF)
- [Crash-DRAW Presentation](#) (PPTX)
- [How to Complete Crash Data Cleanup](#) (PDF)

[Crash Analysis Tools | Ohio Department of Transportation](#)



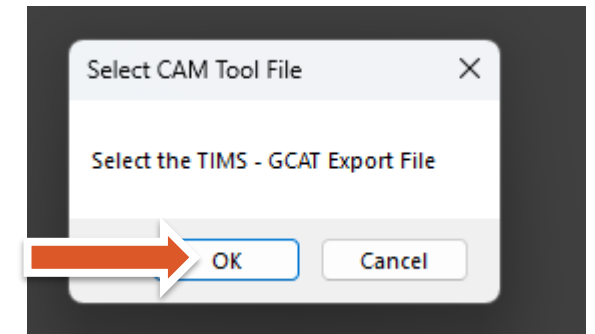
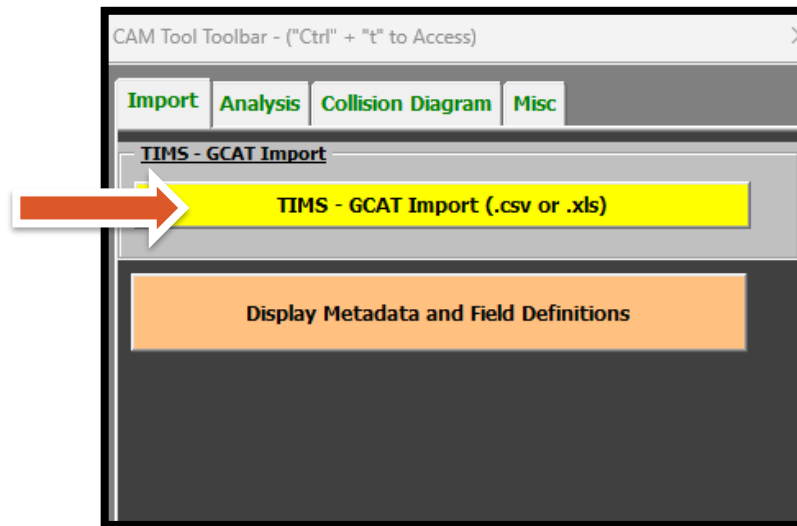
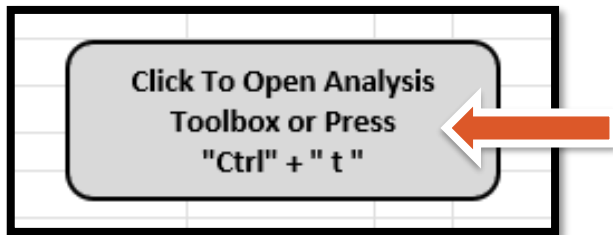
STEP 3

- Open up the downloaded CAM Tool file and make sure to **Enable Content** at the top of the Excel file.



STEP 4

- Select the **Analysis Toolbox** and then **TIMS – GCAT Import**. Select the crash data Excel file you previously downloaded from AASHTOWare Safety.



STEP 5

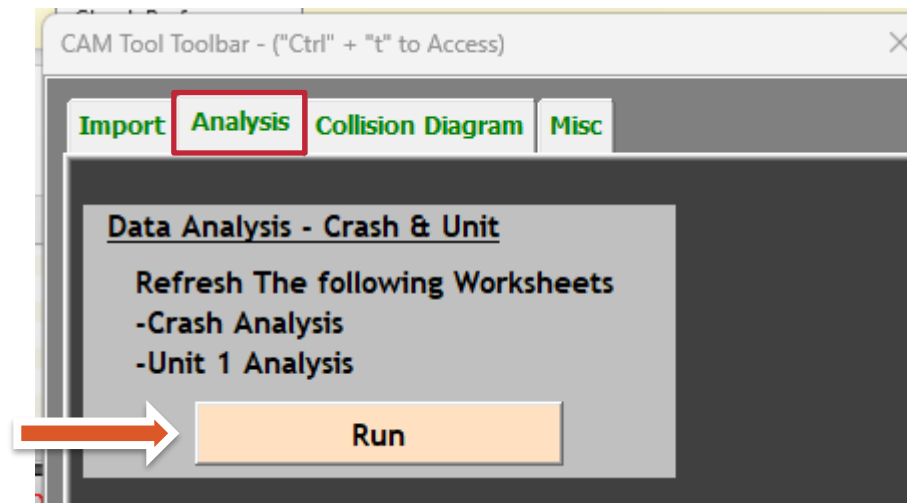
- The **Full Crash Data** sheet will populate with the crash data.
- To open the detailed OH-1 Crash Reports, select the links in Column A.



HYPERLIN	Document	Ye	Local
Crash Report	20203110579	2020	20-165
Crash Report	20186078767	2018	00000
Crash Report	20193282773	2019	1-19-4
Crash Report	20193198101	2019	19-287
Crash Report	20193286632	2019	19-501
Crash Report	20186030789	2018	00000
Crash Report	20203099274	2020	1-120-
Crash Report	20186218106	2018	00000
Crash Report	20203006980	2020	1-20-0
Crash Report	20186050777	2018	00000
Crash Report	20186218087	2018	00000
Crash Report	20205097806	2020	59-058
Crash Report	20203017598	2020	1-20-0
Crash Report	20193056451	2019	1-19-0
Crash Report	20193270631	2019	1-19-4
Crash Report	20203088191	2020	20-143
Crash Report	20205096818	2020	59-057

STEP 6

- Open up the **Analysis Toolbox** again, move to the second tab (**Analysis**), and select **Run** under the **Data Analysis – Crash & Unit** box. The analysis sheets will be updated in the CAM Tool.



STEP 7

- The sheets at the bottom of the file will populate with data analysis in the **Full Crash Data** sheet.
- The **Quick Summary** and **Crash Analysis** sheets provide comprehensive tables and charts of the crash analysis.
- The **Unit 1 Analysis** sheet analyzes information about Unit 1 which is typically the 'at-fault' vehicle.



REMINDER

- Make sure to update the inputs in the **Relative Severity Index** and **Proportions** sheets.

Relative Severity Index Calculation			
ODOT District	3		
Urban Area	Yes		
Freeway	No		
Fill in the Number of Crashes by Crash Type			
Crash Type	Number of Crashes	Crash Type Severity Calc	RSI Multiplier
Unknown	0	\$0	27,128.00
Head On	0	\$0	67,395.00
Rear End	3	\$65,310	21,770.00
Backing	0	\$0	10,385.00
Sideswipe - Meeting	0	\$0	62,150.00
Sideswipe - Passing	2	\$33,482	16,741.00
Angle	8	\$261,200	32,650.00
Parked Vehicle	1	\$18,694	18,694.00
Pedestrian	0	\$0	151,727.00
Animal	0	\$0	10,614.00
Train	0	\$0	75,479.00
Pedalcycles	0	\$0	90,009.00
Other Non-Vehicle	0	\$0	0.00
Fixed Object	3	\$105,066	35,022.00
Other Object	0	\$0	14,631.00
Overturning	0	\$0	81,415.00
Other Non-Collision	0	\$0	36,841.00
Left Turn	0	\$0	34,550.00
Right Turn	0	\$0	18,344.00
Totals	17	\$483,752	
RSI Value =		28,456.00	

Select Site Type	Seg/Rur; 2-lane							
Crash Severity	Site Average		Statewide Average					
	Total (2018-2020)	Total (%)	Total (%)					
Fatal Crash	0	0.00%	0.93%					
Serious Injury Suspected Crash	0	0.00%	4.50%					
Minor Injury Suspected Crash	3	17.65%	14.06%					
Injury Possible Crash	2	11.76%	7.65%					
Property-Damage-Only	12	70.59%	72.86%					
Total	17							
Crashes by Crash Type								
Crash Type	Total (%)		Fatal & All Injury (%)					
	Site Average	Statewide Average	Site Average	Statewide Average				
Unknown	0.00%	0.19%	0.00%	0.12%				
Head On	0.00%	2.86%	0.00%	5.74%				
Rear End	17.65%	10.26%	17.65%	15.40%				
Backing	0.00%	1.12%	0.00%	0.56%				
Sideswipe - Meeting	0.00%	2.30%	0.00%	3.00%				
Sideswipe - Passing	11.76%	3.66%	11.76%	3.92%				
Angle	47.06%	2.36%	47.06%	4.64%				
Parked Vehicle	5.88%	0.81%	5.88%	0.79%				
Pedestrian	0.00%	0.26%	0.00%	0.88%				
Animal	0.00%	33.28%	0.00%	5.60%				
Train	0.00%	0.02%	0.00%	0.03%				
Pedalcycles	0.00%	0.14%	0.00%	0.48%				
Other Non-Vehicle	0.00%	0.01%	0.00%	0.04%				
Fixed Object	17.65%	34.58%	17.65%	47.05%				
Other Object	0.00%	0.92%	0.00%	0.21%				
Falling From Or In Vehicle	0.00%	0.00%	0.00%	0.00%				
Overturning	0.00%	2.75%	0.00%	6.35%				
Other Non-Collision	0.00%	1.30%	0.00%	0.54%				
Left Turn	0.00%	2.66%	0.00%	4.09%				
Right Turn	0.00%	0.52%	0.00%	0.56%				
Crashes by Light Conditions								
	Total (%)		Fatal & All Injury (%)					
<	>	...	Crash Analysis	Unit 1 Analysis	RelativeSeverityIndex	Emphasis Area	Proportions	CrashTree





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CREATING A CRASH DIAGRAM IN THE CAM TOOL

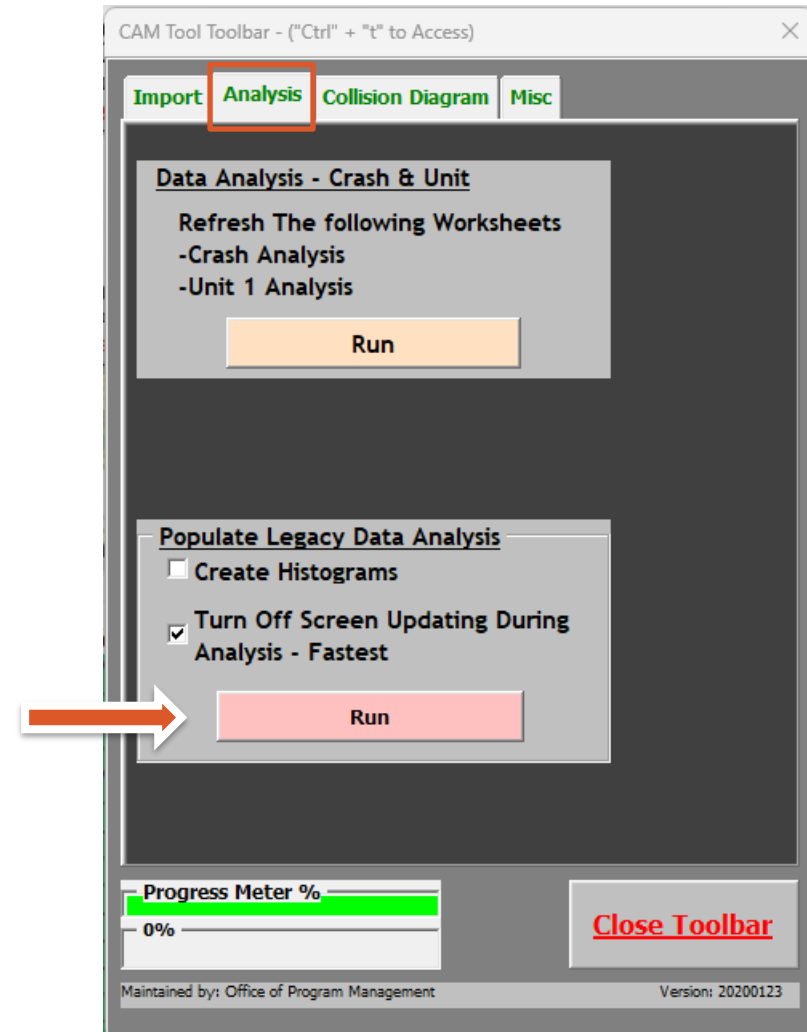
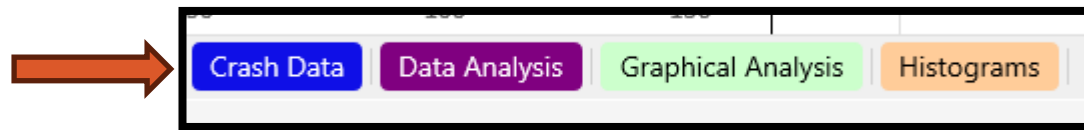
NOTE

- The crash diagram function in the CAM Tool is designed to be an efficient way to plot **intersection** crashes.
- Only include crashes at one intersection in the CAM Tool.
- The best way to create an accurate crash diagram is to open up the crash reports and find the exact locations.



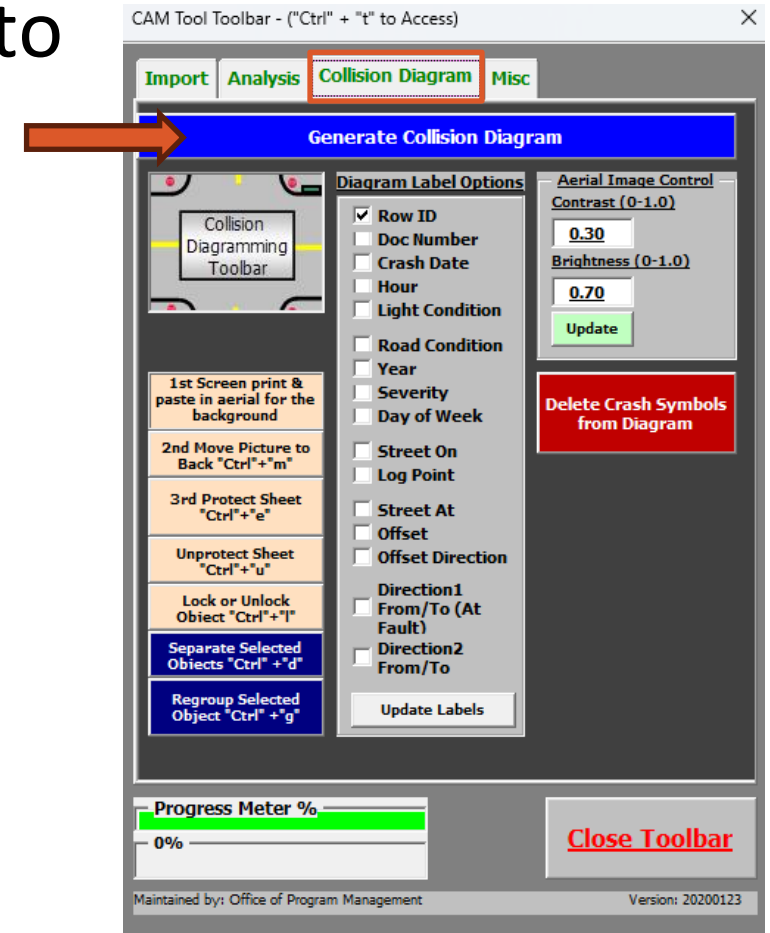
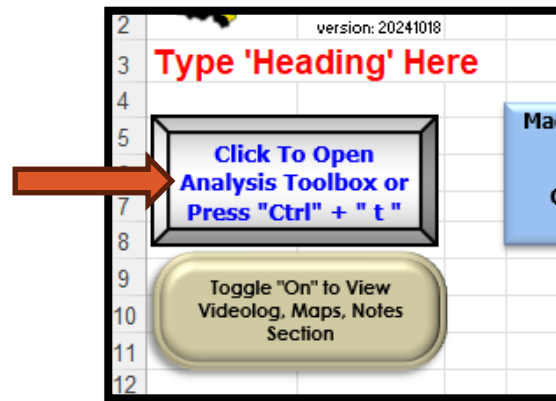
STEP 1

- Go to the second tab of the analysis toolbox and click **Run** in the **Populate Legacy Data Analysis** section.
- Four new sheets will populate:



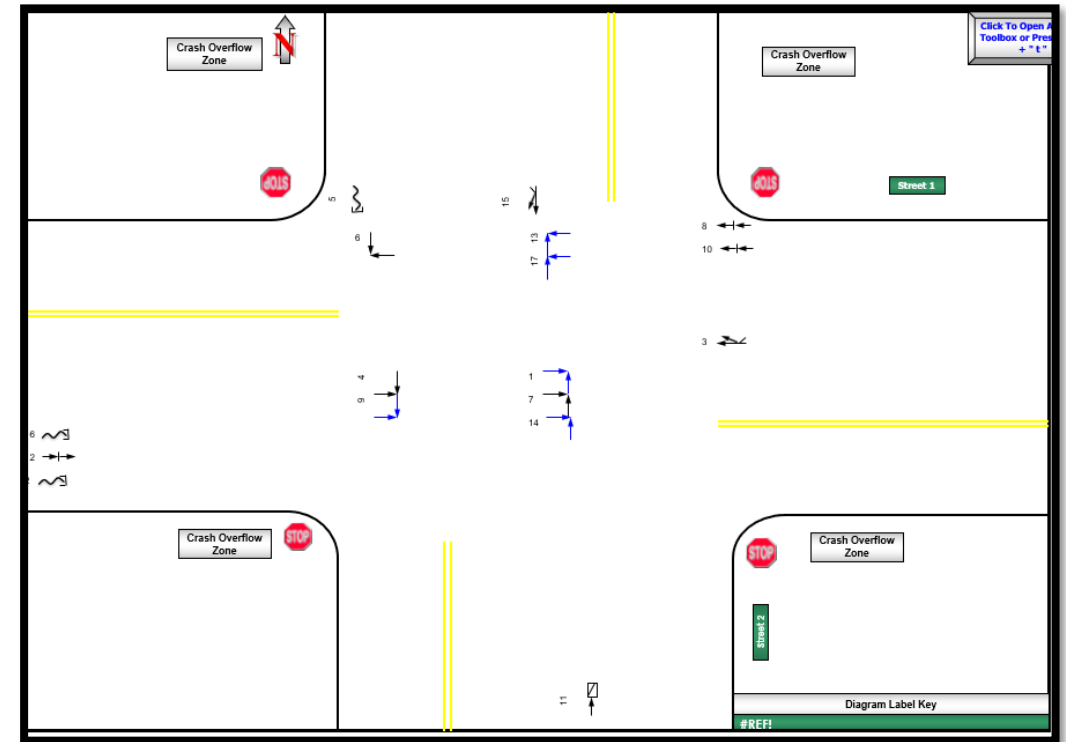
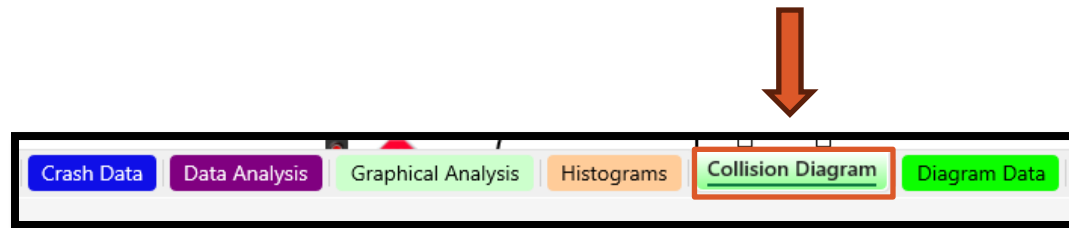
STEP 2

- Go back to the **Analysis Toolbox** and go to the third tab, **Collision Diagram**.
- Check your preferred Diagram Label Options.
- Select the **Generate Collision Diagram** button.



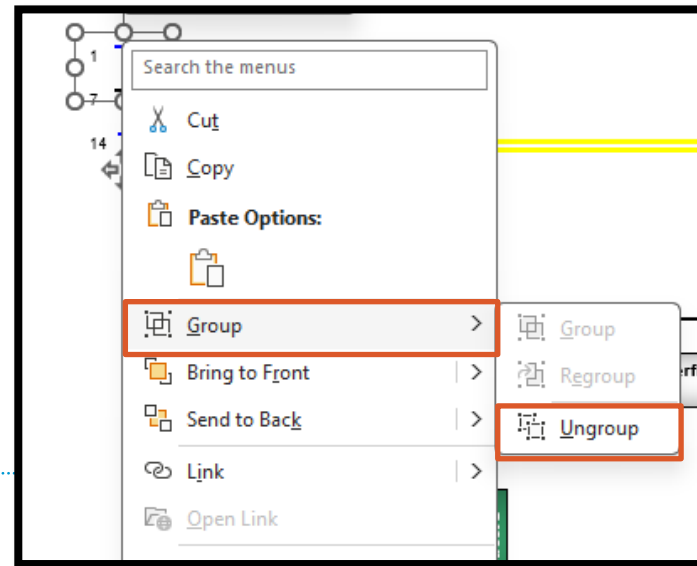
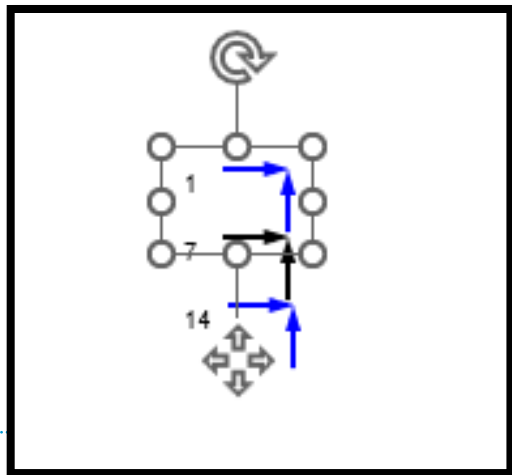
STEP 3

- A diagram will be generated on the **Collision Diagram** sheet with the intersection crashes.



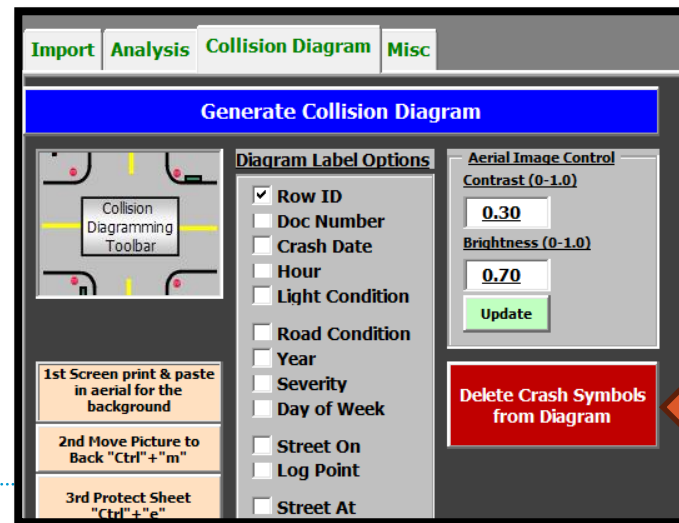
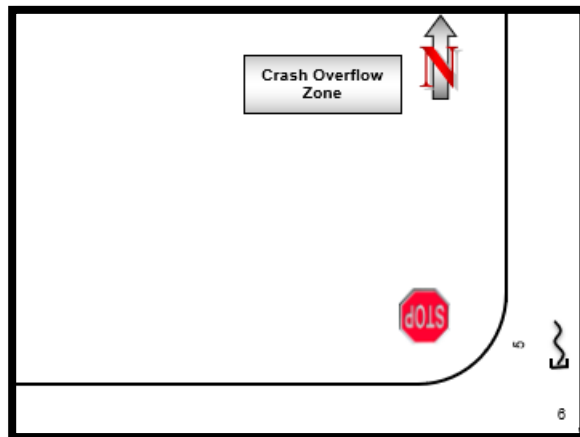
CRASH DIAGRAM NOTES

- Make sure that any changes to the **Crash Type** and **Direction From/To** columns are made prior to analyzing the data and running the collision diagram.
- When first generated, the crashes are located based on the vehicle at fault and the to/from directions.
- If the symbol needs to be rotated it must first be ungrouped by right-clicking on the symbol and selecting ungroup, then the shape can be rotated.



CRASH DIAGRAM NOTES

- If the location has more crashes for a particular movement than the diagram can display it will put the additional crashes in the **Crash Overflow Zone** and allow the user to move the crash manually.
- All symbols can be deleted by clicking the red button **Delete Crash Symbols** from the **Analysis Toolbox**.
- An aerial image can also be included – instructions are located next to the diagram.



ALTERNATIVE CRASH DIAGRAM TOOL

- Also use the **Crash-DRAW Tool** to create crash diagrams
- Find resources here:



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- [How to Complete Crash Data Cleanup](#) (PDF)

[Pages - Safety Resources and Crash Analysis Tools](#)



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THANK YOU

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